# TASK 2: PRODUCT (BOOK) AND CATEGORY MANAGEMENT SYSTEM

## PROJECT DETAILS


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## SCENARIO

The e-commerce bookstore (PALETTE : Pixel & Pages) wants to maintain information about books and their respective categories. Each book belongs to a specific category, and the system should manage book details such as title, ISBN, format (Physical/E-Book), unit price, and stock availability.

## 1.DESIGN PRODUCT (BOOK) AND CATEGORY TABLES

### i. Category Table:

```
SQL> CREATE TABLE Category(
  2    Category_ID VARCHAR2(7),
  3    Category_Name VARCHAR2(100),
  4    Description VARCHAR2(150)
  5  );

Table created.
```

### ii. Product (Book) Table:

```
SQL> CREATE TABLE Book(
  2    Book_ID VARCHAR2(7),
  3    Title VARCHAR2(150),
  4    Category_ID VARCHAR2(7),
  5    Unit_Price NUMBER(9,2),
  6    Format VARCHAR2(20),
  7    ISBN VARCHAR2(20)
  8  );

Table created.
```

The Category and Product (Book) tables have been created successfully. Now, we will define the relationships between these tables and apply necessary constraints to enforce data integrity.

# 2. DEFINE RELATIONSHIPS AND CONSTRAINTS

## i. Relationships:

- One Category can contain many Books (1 -> *).
- Category_ID is the Primary Key in the Category table and acts as a Foreign Key in the Book table to link the two entities.
- Book_ID is the Primary Key in the Book table.

## ii. Constraint Explanations:

- PRIMARY KEY: Uniquely identifies each row in a table (Category_ID and Book_ID).
- FOREIGN KEY: Links the Book table back to the parent Category table (Category_ID).
- NOT NULL: Guarantees that essential fields like Category_Name and Title cannot be left empty.
- CHECK: Validates that Unit_Price is greater than zero and restricts Format strictly to 'Physical' or 'E-Book'.
- UNIQUE: Prevents duplicate values for unique book attributes like ISBN and Category_Name.

## iii. Defining Relationship and Creating Constraints for Product (Book) and Category:

a. Primary Key Constraints

```
SQL> ALTER TABLE Category ADD CONSTRAINT PK_Category PRIMARY KEY (Category_ID);

Table altered.

SQL> ALTER TABLE Book ADD CONSTRAINT PK_Book PRIMARY KEY (Book_ID);

Table altered.
```

b. Foreign Key Constraints

```
SQL> ALTER TABLE Book ADD CONSTRAINT FK_Book_Category
  2  FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID);

Table altered.
```

### iv. Adding Constraints:

a. NOT NULL Constraints

```
SQL> ALTER TABLE Category MODIFY Category_Name VARCHAR2(100) NOT NULL;

Table altered.

SQL> ALTER TABLE Book MODIFY Title VARCHAR2(150) NOT NULL;

Table altered.
```

b. CHECK Constraints

```
SQL> ALTER TABLE Book ADD CONSTRAINT CHK_Price CHECK (Unit_Price > 0);

Table altered.

SQL> ALTER TABLE Book ADD CONSTRAINT CHK_Format CHECK (Format IN ('Physical', 'E-Book'));

Table altered.
```

c. UNIQUE Constraints

```
SQL> ALTER TABLE Category ADD CONSTRAINT UQ_Category_Name UNIQUE (Category_Name);

Table altered.

SQL> ALTER TABLE Book ADD CONSTRAINT UQ_ISBN UNIQUE (ISBN);

Table altered.
```

## 3. INSERT VALUES

### i. Category Table Interactive Insert:

```
SQL> INSERT INTO Category VALUES ('&Category_ID', '&Category_Name', '&Description');
Enter value for category_id: Cat0001
Enter value for category_name: Fiction
Enter value for description: Literary and contemporary fiction novels
old   1: INSERT INTO Category VALUES ('&Category_ID', '&Category_Name', '&Description')
new   1: INSERT INTO Category VALUES ('Cat0001', 'Fiction', 'Literary and contemporary fiction novels')

1 row created.
```

```
SQL> /
Enter value for category_id: Cat0002
Enter value for category_name: Sci-Fi
Enter value for description: Science fiction and space fantasy
old   1: INSERT INTO Category VALUES ('&Category_ID', '&Category_Name', '&Description')
new   1: INSERT INTO Category VALUES ('Cat0002', 'Sci-Fi', 'Science fiction and space fantasy')

1 row created.

SQL> /
Enter value for category_id: Cat0003
Enter value for category_name: Technology
Enter value for description: Computer science and AI technical books
old   1: INSERT INTO Category VALUES ('&Category_ID', '&Category_Name', '&Description')
new   1: INSERT INTO Category VALUES ('Cat0003', 'Technology', 'Computer science and AI technical books')

1 row created.
```

## ii. Product (Book) Table Interactive Insert:

```
SQL> INSERT INTO Book VALUES ('&Book_ID', '&Title', '&Category_ID', &Unit_Price, '&Format', '&ISBN');
Enter value for book_id: B1001
Enter value for title: The Great Gatsby
Enter value for category_id: Cat0001
Enter value for unit_price: 450
Enter value for format: Physical
Enter value for isbn: 9780743273565
old   1: INSERT INTO Book VALUES ('&Book_ID', '&Title', '&Category_ID', &Unit_Price, '&Format', '&ISBN')
new   1: INSERT INTO Book VALUES ('B1001', 'The Great Gatsby', 'Cat0001', 450, 'Physical', '9780743273565')

1 row created.

SQL> /
Enter value for book_id: B1002
Enter value for title: Dune
Enter value for category_id: Cat0002
Enter value for unit_price: 599
Enter value for format: E-Book
Enter value for isbn: 9780441172719
old   1: INSERT INTO Book VALUES ('&Book_ID', '&Title', '&Category_ID', &Unit_Price, '&Format', '&ISBN')
new   1: INSERT INTO Book VALUES ('B1002', 'Dune', 'Cat0002', 599, 'E-Book', '9780441172719')

1 row created.

SQL> /
Enter value for book_id: B1003
Enter value for title: Foundation
Enter value for category_id: Cat0002
Enter value for unit_price: 650
Enter value for format: Physical
Enter value for isbn: 9780553293357
old   1: INSERT INTO Book VALUES ('&Book_ID', '&Title', '&Category_ID', &Unit_Price, '&Format', '&ISBN')
new   1: INSERT INTO Book VALUES ('B1003', 'Foundation', 'Cat0002', 650, 'Physical', '9780553293357')

1 row created.
```

```
SQL> /
Enter value for book_id: B1004
Enter value for title: Designing Data-Intensive Applications
Enter value for category_id: Cat0003
Enter value for unit_price: 1250
Enter value for format: Physical
Enter value for isbn: 9781449373320
old   1: INSERT INTO Book VALUES ('&Book_ID', '&Title', '&Category_ID', &Unit_Price, '&Format', '&ISBN')
new   1: INSERT INTO Book VALUES ('B1004', 'Designing Data-Intensive Applications', 'Cat0003', 1250, 'Physical', '9781449373320')

1 row created.
```

Like this insert few more rows.

## iii. Display Tables:

a. Category Table

```
SQL> SELECT * FROM Category;

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
DESCRIPTION
--------------------------------------------------------------------------------
Cat0001
Fiction
Literary and contemporary fiction novels

Cat0002
Sci-Fi
Science fiction and space fantasy

CATEGOR
-------
CATEGORY_NAME
--------------------------------------------------------------------------------
DESCRIPTION
--------------------------------------------------------------------------------

Cat0003
Technology
Computer science and AI technical books
```

b. Product (Book) Table

```
SQL> SELECT * FROM Book;

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
CATEGOR UNIT_PRICE FORMAT             ISBN
------- ---------- ------------------ --------------------
B1001
The Great Gatsby
Cat0001        450 Physical           9780743273565

B1002
Dune
Cat0002        599 E-Book             9780441172719

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
CATEGOR UNIT_PRICE FORMAT             ISBN
------- ---------- ------------------ --------------------

B1003
Foundation
Cat0002        650 Physical           9780553293357

B1004
Designing Data-Intensive Applications

BOOK_ID
-------
TITLE
--------------------------------------------------------------------------------
CATEGOR UNIT_PRICE FORMAT             ISBN
------- ---------- ------------------ --------------------
Cat0003       1250 Physical           9781449373320
```

# 4. CRUD OPERATIONS

## i. Insert Record:

```
SQL> INSERT INTO Book (Book_ID, Title, Category_ID, Unit_Price, Format, ISBN)
  2  VALUES ('B1005', 'Thulasimadam', 'Cat0001', 250, 'Physical', '9788123456789');

1 row created.
```

## ii. Update Price:

```
SQL> UPDATE Book SET Unit_Price = 499 WHERE Title = 'The Great Gatsby';

1 row updated.
```

## iii. Delete a Product (Book):

```
SQL> DELETE FROM Book WHERE Title = 'Thulasimadam';

1 row deleted.
```

# 5. GENERATE CATEGORY-WISE PRODUCT (BOOK) REPORTS

## i. Display all products (books) under each category:

```
SQL> SELECT c.Category_Name, b.Title, b.Unit_Price, b.Format
  2  FROM Book b JOIN Category c ON b.Category_ID = c.Category_ID;

CATEGORY_NAME
--------------------------------------------------------------------------------
TITLE
--------------------------------------------------------------------------------
UNIT_PRICE FORMAT
---------- --------------------
Fiction
The Great Gatsby
       499 Physical

Sci-Fi
Dune
       599 E-Book

CATEGORY_NAME
--------------------------------------------------------------------------------
TITLE
--------------------------------------------------------------------------------
UNIT_PRICE FORMAT
---------- --------------------

Sci-Fi
Foundation
       650 Physical

Technology
Designing Data-Intensive Applications
```

```
CATEGORY_NAME
--------------------------------------------------------------------------------
TITLE
--------------------------------------------------------------------------------
UNIT_PRICE FORMAT
---------- --------------------
      1250 Physical
```

## ii. Count number of Products (Books) in each Category:

```
SQL> SELECT c.Category_Name, COUNT(b.Book_ID) AS Book_Count
  2  FROM Book b JOIN Category c ON b.Category_ID = c.Category_ID
  3  GROUP BY c.Category_Name;

CATEGORY_NAME
--------------------------------------------------------------------------------
BOOK_COUNT
----------
Fiction
         1

Sci-Fi
         2

Technology
         1
```

## iii. Find highest-priced Product (Book) in each Category:

```
SQL> SELECT c.Category_Name, b.Title, b.Unit_Price AS Highest_Price
  2  FROM Book b JOIN Category c ON b.Category_ID = c.Category_ID
  3  WHERE b.Unit_Price = (
  4      SELECT MAX(b2.Unit_Price)
  5      FROM Book b2
  6      WHERE b2.Category_ID = b.Category_ID
  7  );

CATEGORY_NAME
--------------------------------------------------------------------------------
TITLE
--------------------------------------------------------------------------------
HIGHEST_PRICE
-------------
Fiction
The Great Gatsby
          499

Sci-Fi
Foundation
          650

CATEGORY_NAME
--------------------------------------------------------------------------------
TITLE
--------------------------------------------------------------------------------
HIGHEST_PRICE
-------------

Technology
Designing Data-Intensive Applications
         1250
```

## iv. Display Categories having more than 1 Product (Book):

```
SQL> SELECT c.Category_Name
  2  FROM Book b JOIN Category c ON b.Category_ID = c.Category_ID
  3  GROUP BY c.Category_Name
  4  HAVING COUNT(b.Book_ID) > 1;

CATEGORY_NAME
--------------------------------------------------------------------------------
Sci-Fi
```

## v. Find average Product (Book) price Category-wise:

```
SQL> SELECT c.Category_Name, AVG(b.Unit_Price) AS Avg_Price
  2  FROM Book b JOIN Category c ON b.Category_ID = c.Category_ID
  3  GROUP BY c.Category_Name;

CATEGORY_NAME
--------------------------------------------------------------------------------
 AVG_PRICE
----------
Fiction
       499

Sci-Fi
     624.5

Technology
      1250
```